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VTV-FM: Flow Matching through Variational Terminal-Velocity Closure

Minimum-Acceleration Energy Provides Closed-Form Terminal Velocities for Second-Order Bridges, Reducing CIFAR-10 FID to 2.83 at 50 NFE

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First-order flow matching uses straight-line bridges between noise and data; second-order dynamics would allow curved, lower-NFE paths, but static datasets carry no terminal-velocity information. VTV-FM derives the missing terminal velocity via a minimum-acceleration energy principle, yielding a closed-form cubic bridge that reduces CIFAR-10 FID from 3.11 to 2.83 at 50 NFE and cuts the FID at 5 NFE from 9.48 to 7.21.

AT A GLANCE

Problem: Second-order flow matching needs terminal velocities at data points, which static image datasets do not provide.

Why it matters: Curved paths reduce the number of function evaluations needed during sampling, lowering inference cost.

Method: Derive terminal velocity via minimum-acceleration energy; fit cubic Hermite bridges; train velocity and acceleration networks.

Result: CIFAR-10 FID 2.83 at 50 NFE (vs. 3.11 linear FM); FID 7.21 at 5 NFE (vs. 9.48 linear FM).

Evidence: Low-dimensional datasets, PDE physical fields, CIFAR-10; ablations on σ_v, OT cost, acceleration head.

The Big Picture

Flow matching (FM) learns a velocity field that transports a noise distribution p_0 to a data distribution p_1 along a continuous path x_t. The dominant approach uses **first-order linear bridges**:

$$x_t = (1-t)x_0 + tx_1, \quad v(x_t, t) = x_1 - x_0$$

These are analytically convenient but geometrically naive: real data distributions are rarely connected by straight lines. Second-order bridges model curvature and allow faster sampling at fewer network evaluations (NFE).

The obstacle to second-order FM is that learning a cubic or higher-order bridge requires terminal velocities $\dot{x}_1$ at data endpoints. Static image datasets do not supply these. VTV-FM provides the first principled solution.

Why Existing Approaches Fall Short

Linear FM (Lipman et al. 2022; Liu et al. 2022): Straight-line paths; many NFE required to follow curved manifolds.

Stochastic interpolants / DDPM: Noisy paths; good for diversity but not for deterministic low-NFE sampling.

High-order FM (Ho et al. 2023): Proposes higher-order bridges but requires paired velocity annotations $(\dot{x}_0, \dot{x}_1)$—unavailable for static datasets.

OT-FM (Tong et al. 2023): Globally optimal pairings but linear bridges; curvature not modeled.

VTV-FM: Derives $\dot{x}_1^*$ from minimum energy without velocity annotations, enabling second-order FM for any static dataset.

Core Mechanism: Minimum-Acceleration Energy

For a path γ_t connecting x_0 and x_1 with initial velocity $\dot{x}_0$, the minimum-acceleration energy functional is:

$$E[\gamma] = \int_0^T \|\ddot{\gamma}_t\|^2 \, dt$$

Minimising over cubic Hermite paths subject to $\gamma_0 = x_0$, $\gamma_T = x_1$, $\dot{\gamma}_0 = \dot{x}_0$ yields a closed-form terminal velocity:

$$\dot{x}_1^* = \frac{6(x_1 - x_0)}{T} - 3\dot{x}_0$$

This requires no annotation: $\dot{x}_0$ is sampled from a Gaussian prior $\mathcal{N}(0, \sigma_v^2 I)$, and $\dot{x}_1^*$ is computed analytically.

Technical Formulation

The cubic Hermite bridge parameterised by $(x_0, \dot{x}_0, x_1, \dot{x}_1^*)$:

$$\gamma_t = h_{00}(t)x_0 + h_{10}(t)\dot{x}_0 + h_{01}(t)x_1 + h_{11}(t)\dot{x}_1^*$$

where $h_{00}, h_{10}, h_{01}, h_{11}$ are the standard Hermite basis polynomials.

Training objectives:

$$\mathcal{L}_v(\theta) = \mathbb{E}_{t,x_0,x_1,\dot{x}_0}\left[\|v_\theta(\gamma_t, t) - \dot{\gamma}_t\|^2\right]$$

$$\mathcal{L}_a(\phi) = \mathbb{E}_{t,x_0,x_1,\dot{x}_0}\left[\|a_\phi(\gamma_t, t) - \ddot{\gamma}_t\|^2\right]$$

The extended OT coupling cost is:

$$C_{\text{VTV}}(x_0, x_1) = \frac{2\|x_1 - x_0\|^2}{T^2} - \frac{2\langle \dot{x}_0, x_1 - x_0\rangle}{T} + \frac{\|\dot{x}_0\|^2 T}{3}$$

reducing to the standard L^2 cost when $\dot{x}_0$=0.

Learning or Inference Procedure

1. Sample (x_0, x_1) from the coupling; draw $\dot{x}_0 \sim \mathcal{N}(0, \sigma_v^2 I)$.
2. Compute $\dot{x}_1^*$ analytically; evaluate $\gamma_t, \dot{\gamma}_t, \ddot{\gamma}_t$.
3. Train v_θ and a_ϕ with regression losses $\mathcal{L}_v, \mathcal{L}_a$.
4. At inference, integrate the second-order ODE using velocity-Verlet or RK4 with as few as 5 NFE.

What the Guarantee Says

The cubic bridge γ_t with $\dot{x}_1^*$ is the unique minimum-acceleration path under the boundary conditions. The curvature $\|\ddot{\gamma}_t\|$ is bounded by $\mathcal{O}(\sigma_v)$, and decreases to zero as $\sigma_v \to 0$, recovering the linear bridge in that limit. The VTV coupling cost reduces to OT-FM's cost when $\dot{x}_0$=0.

Experimental Findings

- **Low-dimensional:** 42% lower mean curvature and 31% lower Wasserstein-2 at 5 NFE vs. linear FM on 2D distributions.
- **Navier-Stokes PDE fields:** RMSE 0.042 vs. 0.067 (linear FM) at 5 NFE.
- **CIFAR-10:** FID **2.83** at 50 NFE vs. 3.11 (linear FM), 3.04 (rectified flow). FID **7.21** at 5 NFE vs. 9.48 (linear FM).
- Training overhead: ≈15% from acceleration head; inference unchanged if only v_θ is used.

Ablations and Interpretation

$\dot{x}_0$=0**:** Cubic paths degenerate to symmetric splines; performance slightly below full VTV-FM with sampled $\dot{x}_0$.

σ_v **sensitivity:** CIFAR-10 FID stable for $\sigma_v \in [0.5, 2.0]$; too large (overcurved paths) or too small (near-linear, wasting the second-order structure) both hurt.

Acceleration head: Using a_ϕ with a second-order ODE solver reduces NFE by 40% at FID 3.0 vs. first-order integration of v_θ alone.

OT coupling: VTV coupling reduces FID by 0.22 on CIFAR-10 vs. standard L^2 cost, confirming the extended cost captures anisotropic transport structure.

Reference

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